

SILUNI WIJESINGHE

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PROFILE

B.Sc. (Hons.) Information Technology graduate **(University of Moratuwa**, coursework completed, degree conferral December 2026) with six months of industry experience developing and maintaining **enterprise ERP solutions in production**. Hands-on experience across **frontend, backend, and database development** using **React.js**, **ASP.NET Core**, **Spring Boot**, and **SQL Server**, with skills in **RESTful APIs**, **ORM**, and **SQL optimization**. Strong foundation in **software engineering and problem-solving**, with additional experience in **machine learning and computer vision** through academic research.

EXPERIENCE

Intern Full Stack Software Engineer Feb 2025 - Aug 2025
Agrithmics (Pvt) Ltd, Sri Lanka

- **Enterprise ERP - Full-Stack Feature Delivery:** Built full-stack features across 4 international ERP platforms - Ispahani, Lankem, Hayleys, and HNB Farmer System using C#/.NET and React, spanning procurement, payroll, inventory, and manufacturing modules.
- **Feature Engineering:** Developed **10+ screens** including Quotation Management screen with item-wise multi-vendor selection logic and automated comparison scoring, reducing manual procurement decision time by consolidating vendor evaluation onto a single screen; implemented Excel export for bulk quote downloads.
- **Background Process Debugging:** Traced and fixed a critical execution bug in the Sick Leave Scheduler causing incorrect payroll processing in production; resolved by analysing scheduling logic and service execution flow within a single sprint.
- **Reporting & SQL Optimisation:** Delivered **dynamic reporting modules** including a Maternity Leave Weekly Payment Report with automated PDF generation, configurable date-range validation, and per-column dynamic generation, reducing manual HR report preparation time.
- **Codebase Quality & Performance:** Refactored **5+ legacy query modules** following senior code review; replaced N+1 query patterns with optimised SQL joins, measurably reducing report load times on data sets exceeding **10,000 records**.
- **Bug Fixing & Code Quality:** Resolved production bugs across inventory, payroll, attendance, and manufacturing screens; refactored codebase following supervisor feedback to improve query efficiency and reduce redundancy.
- **DevOps & Agile Delivery:** Managed sprint tasks via Azure DevOps - Git branching, pull requests with peer review, unit tests, code reviews.

SOFTWARE PROJECTS

CentralSync - Cross-Platform Inventory Management System [Frontend] [Backend] [Mobile]
Rootcode Labs | React.js, React Native, Spring Boot, MySQL

- Developed full-stack resource management and order initiation features, implementing inventory CRUD and ordering workflows using React.js and Spring Boot.
- Built an item usage analysis and reporting module with Spring Boot backend logic and React.js visualizations, enabling data-driven analysis of inventory consumption.
- Implemented autocomplete-based search and filtering and contributed to requirements analysis, UML design, and frontend/backend integration.

Pathwise - AI-Powered Career Guidance Platform [Frontend] [Backend]
Idealize 2025 | Next.js, Spring Boot, MySQL, Generative AI

- Developed coach-session booking system with conflict detection and scheduling logic.
- Built real-time WebSocket-based messaging module with persistence.
- Integrated full-stack flow from Next.js frontend to Spring Boot APIs and MySQL database.

DevAssist - AI Codebase Assistant for GitHub Repositories [GitHub]
Personal Project | Java 21, Spring Boot, React, PostgreSQL, JGit, Docker, LLM Tool Calling

- Built a full-stack assistant that gives an LLM agent tool-based access (code search, file reading, Git history) to imported GitHub repositories, so answers come from the actual code instead of pasted snippets, with JGit ingestion and PostgreSQL storage.

- Secured it with Spring Security and JWT plus per-user data isolation (JUnit-tested), and added persistent chat history with conversation context and a repository re-sync that flags outdated answers.
- Deployed a Dockerised Spring Boot + React app on Render with a Neon PostgreSQL database

QA Automation Suite - QA Training App [GitHub]

Group Project | Cucumber, Cypress, JavaScript, Maven

- Built BDD-driven UI and API test automation for a Spring Boot app using Cucumber and Cypress, covering Category and Sales modules and role-based access control for Admin/User accounts.

RESEARCH PROJECTS

Machine Vision System for Morphologically Similar Ayurvedic Compound-Leaf Species Identification & Health Assessment [GitHub]

Final Year Research Project | Python, OpenCV, scikit-learn, TensorFlow, FastAPI

- Developed an end-to-end **computer vision pipeline** for identifying 12 morphologically similar Ayurvedic plant species from compound-leaf images, covering image preprocessing, feature engineering, model training, and evaluation.
- Engineered 130+ handcrafted shape, colour, texture, and vein features and developed a soft-voting ensemble using Random Forest, SVM-RBF, and HistGradientBoosting, achieving 93.7% F1-macro on a sealed 240-image test set.
- Developed a two-stage leaf health assessment system for healthy/unhealthy classification and severity assessment, achieving 0.885 F1 score and benchmarking handcrafted features against a MobileNetV2 deep-learning baseline through ablation experiments.

Differential Privacy in Machine Learning for Social Media Recommender Systems

Level 3 Independent Study Review Paper

- Reviewed differential privacy in ML recommender systems (Local DP, DP-SGD, HE/SMPC hybrids), analysing the privacy-utility trade-off and geo-social graph privacy in POI and social recommendation.

TECHNICAL SKILLS

Languages : Java, C#, JavaScript, C++, Python
Frontend : React, Next.js, React Native, Tailwind CSS, HTML/CSS
Backend : Spring Boot, .NET Framework,Spring Security, JWT ,RESTful APIs, Entity Framework
Database : MySQL, Microsoft SQL Server, PostgreSQL
DevOps & Tools : Azure DevOps, Docker, Git, GitHub, Postman, Agile/Scrum

EDUCATION

BSc. (Hons) in Information Technology **2022 – 2026**
University of Moratuwa | GPA: 3.75/4.00 (First Class)
Dean's List: L1S2 (SGPA 3.92) · L2S1 (SGPA 4.00) · L2S2 (SGPA 3.96) · L4S1 (SGPA 3.88) · L4S2 (SGPA 3.81)

G.C.E. Advanced Level - Physical Science **2020**
Anula Vidyalaya, Nugegoda | Z-score: 1.7467 | 2As & 1B

G.C.E. Ordinary Level **2017**
Ananda Balika Vidyalaya, Kotte | 9 A's

HACKATHONS

Idealize (2025) | Inter-University Designathon | Semi-Finalist - Team BitBalance
HackElite 1.0 (2024) | Inter-University Ideathon | Semi-Finalist - Team VivalaCodes
InspiHer-Tech V2.0 (2023) | Programming Hackathon | Semi-Finalist - Team VivalaCodes
Coderush (2023) | Programming Hackathon
Devthon (2023) | Programming Hackathon

REFERENCES

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